Matching Evidence to Decisions in Prolonged Crises

Can we say beforehand how far a finding travels, and for which decision?

Aaron Scherf • Columbia University | Sustainable Development (SIPA)

> **Example of the decision we study (hypothetical).** After repeated drought, should a programme extend short-term cash transfers or link households to a long-term safety net? These choices sit in the **humanitarian–development nexus**: coordinating emergency aid with long-term development.

Why these decisions matter: crises are large and last for years

- Fragile and conflict-affected settings hold a growing share of the world's extreme poor (World Bank, 2025 projection)
- About seven in ten refugees are in protracted situations, displaced for years rather than months (UNHCR, 2025)
- About 239 million people need humanitarian aid (OCHA, 2026)

Who decides, what they decide, and why discretion matters

Topic of study: the **humanitarian–development nexus**, the established term for coordinating life-saving aid with longer-term development when shocks repeat or crises last for years.

Immediate response

Example: who receives emergency food next month

Who shapes it

Field planners

Where most decision studies sit

Nexus design: this agenda

Example: how to sequence and link cash, safety nets and livelihood support over several years

Who shapes it

Field planners · Headquarters · Donors and governments · Evidence funders · Researchers

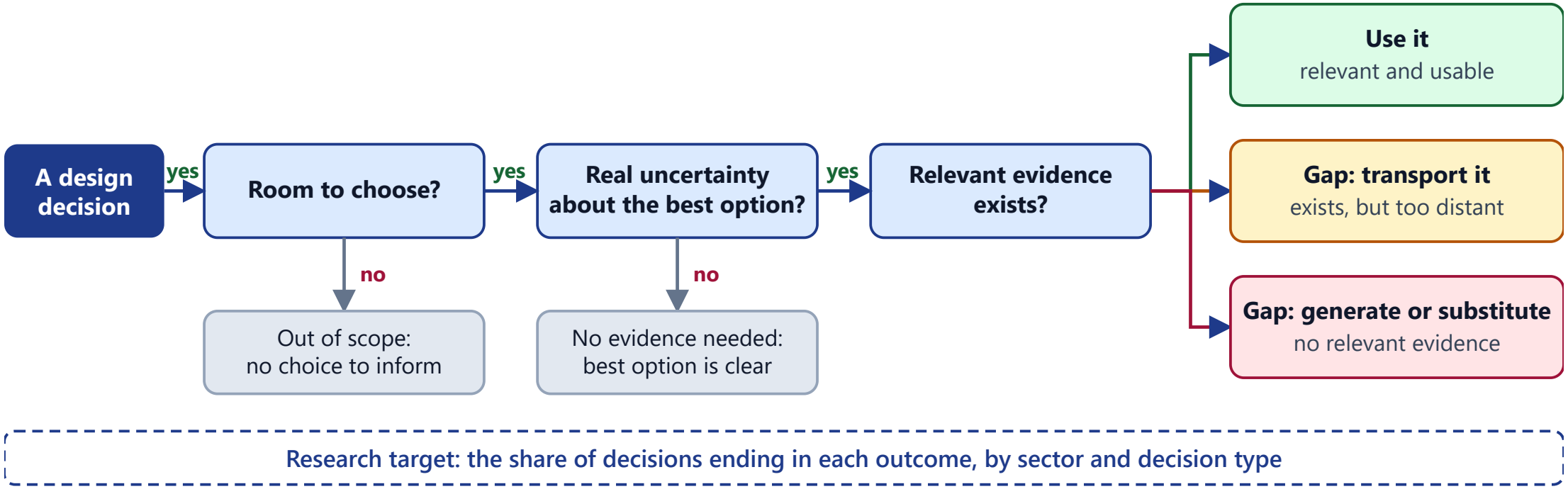
Rules · budget · mandate · norms

Planner discretion: the room left to choose

Rules · budget · mandate · norms

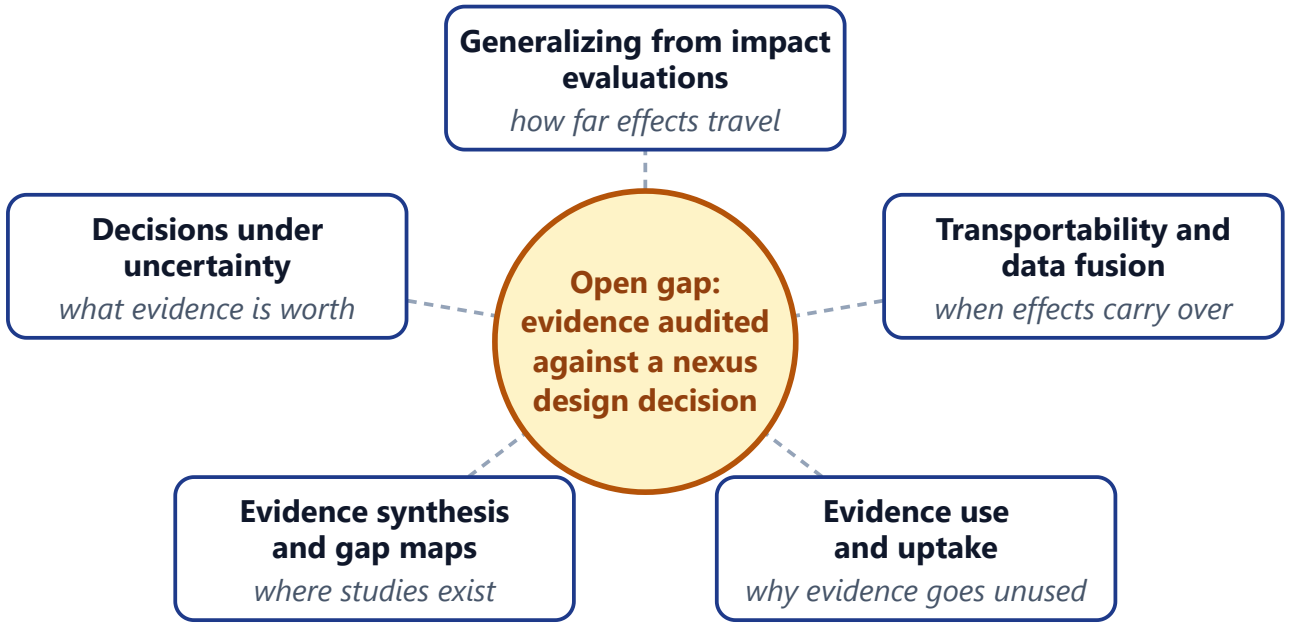
- Evidence can only change choices that are open: in the example, extending cash or linking to a safety net is a real choice only if the rules and budget allow both
- So we **measure discretion** decision by decision instead of assuming it

Evidence is worth seeking only if it could change the choice



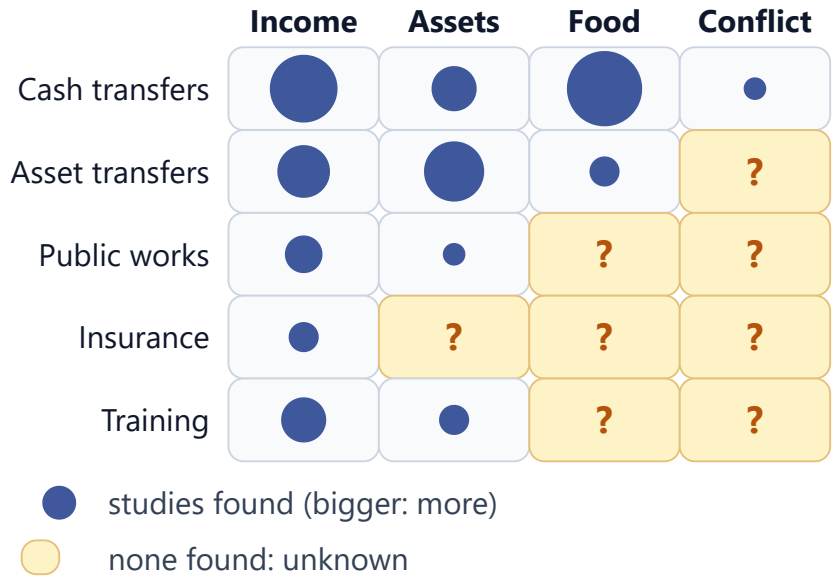
- **Value of information** comes from health economics: more evidence is worth getting only if it could change which option is chosen. We apply it to programme design
- Relevance has two parts: **distance** (will it hold here?) and **actionability** (can this planner implement it?)
- A real case: the business case for Phase 2 of Kenya's Hunger Safety Net Programme (HSNP, a cash transfer programme) weighed seven expansion options. The evidence it reviewed did not clearly separate them

Where this sits in the literature



3ie evidence gap map (schematic)

We extend it with context filters:
shock, crisis phase, delivery



- Nearest humanitarian work found: IDinsight (value of information for commissioning an evaluation), ALNAP (immediate-response decisions), Elrha (barriers). None audits existing evidence against a nexus design decision. A **3ie evidence gap map** is a matrix of interventions by outcomes showing where studies exist
- **Why start in crises:** the hard case has the least evidence and the most constraint, so a method that works here should transfer to fields with more and closer evidence

Two Kenyan cash studies disagree; one programme travelled to conflict

Same country, two answers

Egger et al.
randomized trial

PAYMENT



One-time cash transfer

WHO
Poor rural households

WHAT IT MEASURES
Spillovers to the local economy

?
defined differently

HSNP Phase 2 evaluation
local-economy model

PAYMENT



Unconditional cash every two months

WHO
Chronically poor households in four arid northern counties

WHAT IT MEASURES
Income created locally, per shilling transferred

Evidence that travelled

Bedoya et al., Afghanistan
graduation programme

PROGRAMME PACKAGE

- Livestock asset
- Cash stipend
- Skills training
- Coaching

Stable settings:
success shown first

↓

Conflict setting:
tested, large effects reported

Pre-registered; no replication data found

The Kenyan studies report different quantities (a spending multiplier versus income per shilling), and why their results differ is not established. Both are on our replication list. *Travelled* means the same programme was evaluated in a conflict setting; it is not a transport test.

Evidence–decision mismatch for humanitarian–development planners

The problem. Evidence is a non-market good: no price tells funders which studies are needed, so research attention can be misallocated while nexus planners (those who design aid programmes) still lack evidence they can use. The mismatch is one of relevance.

Discretion and value of evidence

How much room do nexus planners have to choose, and how useful do they find the evidence available to them?

Decision mapping

What informs their decisions today, and what keeps relevant evidence from reaching them: hard to find, too much of it, or not relevant to their setting?

Evidence transport

Which barriers to using existing evidence can be reduced, and how well can we predict an intervention's effect in a new setting?

▼

Primary goal: better-designed interventions

Checking predictions is a means to this, not the goal itself

- More efficient allocation of research funding
- More targeted evidence generation by development organizations
- An open-source toolkit that generalizes to other uses

Discretion and value of evidence: what we measure first

Asked of every decision

- **Discretion left?** after rules and budget
- **Existing information** can it separate the options?
- **Timeline** does it allow more evidence?
- **Cost of being wrong** who bears it; reversible?

First: desk audit

documents only, no human subjects

- Decisions made
- Constraints
- Evidence used
- Outcomes

Also: did decision-makers seek outside support?

then

Later: partner funnel

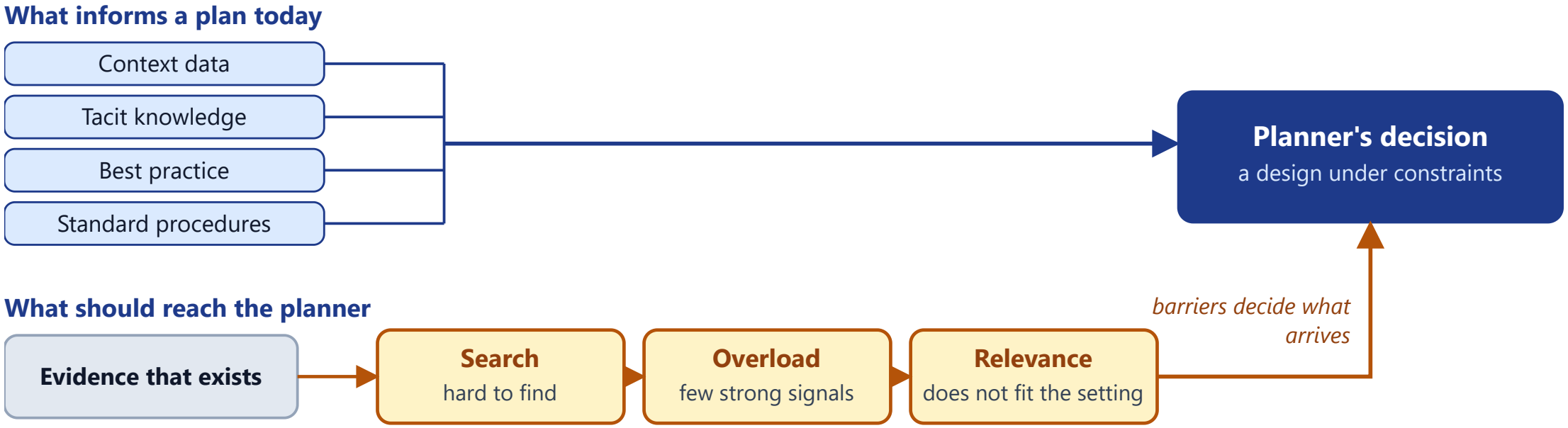
refusals are data



- In scope: headquarters versus field constraints; where evidence generation should be allocated

Still open: the taxonomy of decisions (immediate response, long-term planning, operational, design), and the headquarters/field relationship.

Decision mapping: what informs decisions today



- The unit of study is the **assumptions** behind a decision, in causal maps, scenarios and log-frames (the logic tables programmes write), stated or not
- **Feeds evidence transport:** the causal chain, blockers and log-frame assumptions say which links a source study must be checked on; the planner reviews them at each staged deliverable
- Eliciting causal maps from practitioners is the established field of **systems mapping**, usually qualitative. Linking it to causal graphs for quantitative transport is the untested part

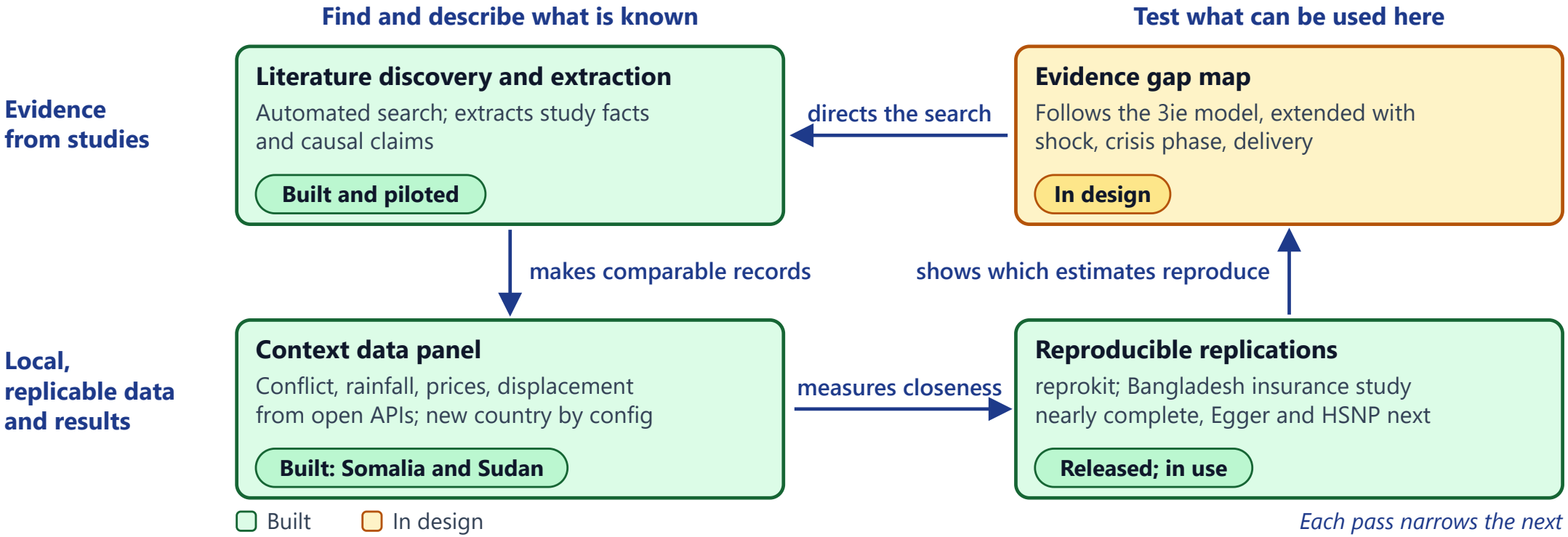
Evidence transport: how far does a finding travel? What counts as comparable

- We predict an intervention's effect in a new setting (the **target**) from studies done elsewhere (the **source**), to set a realistic target, then check it against the measured outcome
- Source and target are compared along named dimensions: spatial, temporal, institutional, market, treatment version, population, interference
- **Comparable is a profile, not one score.** The measured similarity and the planner's own rating must both clear the planner's thresholds, and a disagreement between them is flagged. Thresholds decide which prediction type is defensible (next slides)
- Idea for choosing cases: an insecure region may resemble an insecure region elsewhere more than the rest of its own country



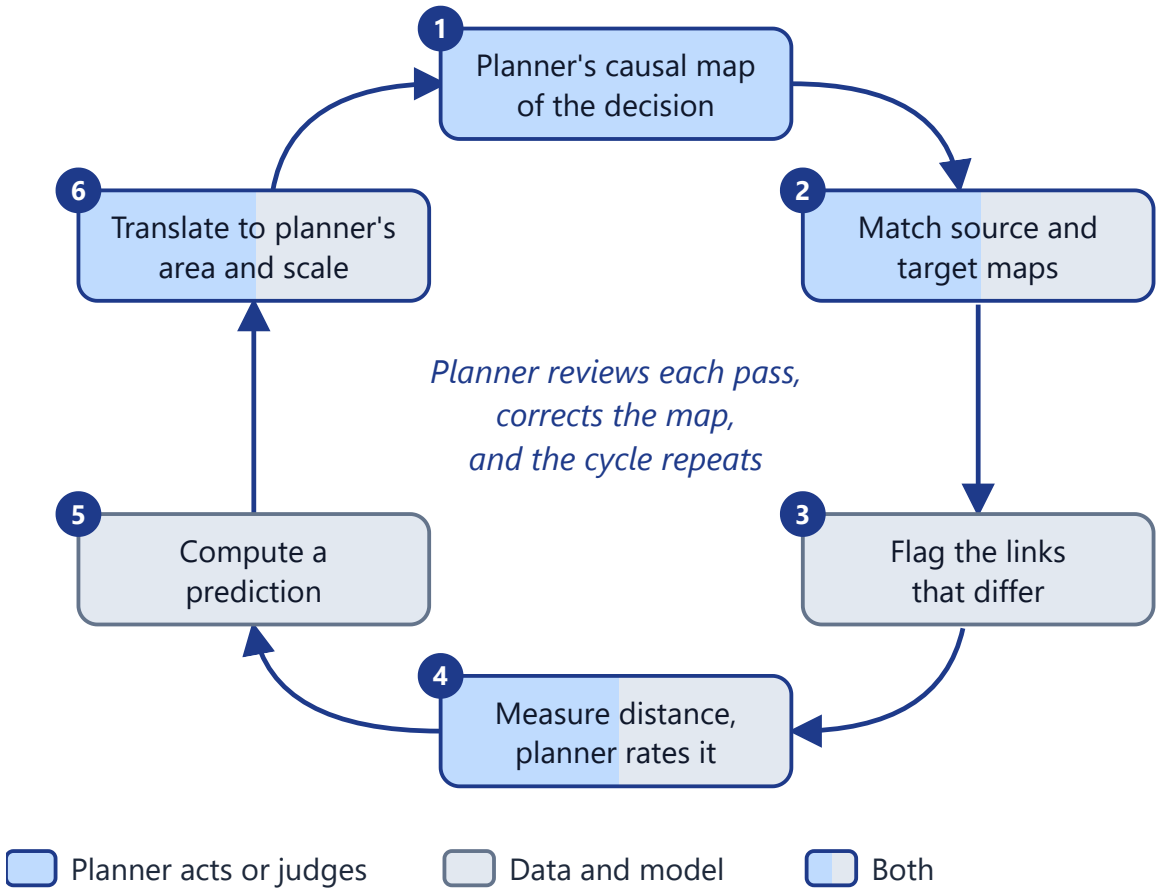
The scale and the values are illustrative. More than one expert rating is ideal, subject to availability.

The pipeline already under construction



- `reprokit`, our reproducibility toolkit, fingerprints files (hashes) and rebuilds results automatically (continuous integration), so anyone can re-run a result
- Replication is expected to be fast when geodata are public or not needed (not yet timed); slower if locations must be requested privately
- A **proof of concept for a re-runnable, open-source pipeline**, validated in the dissertation on one applied case

The transport model: built with the decision-maker

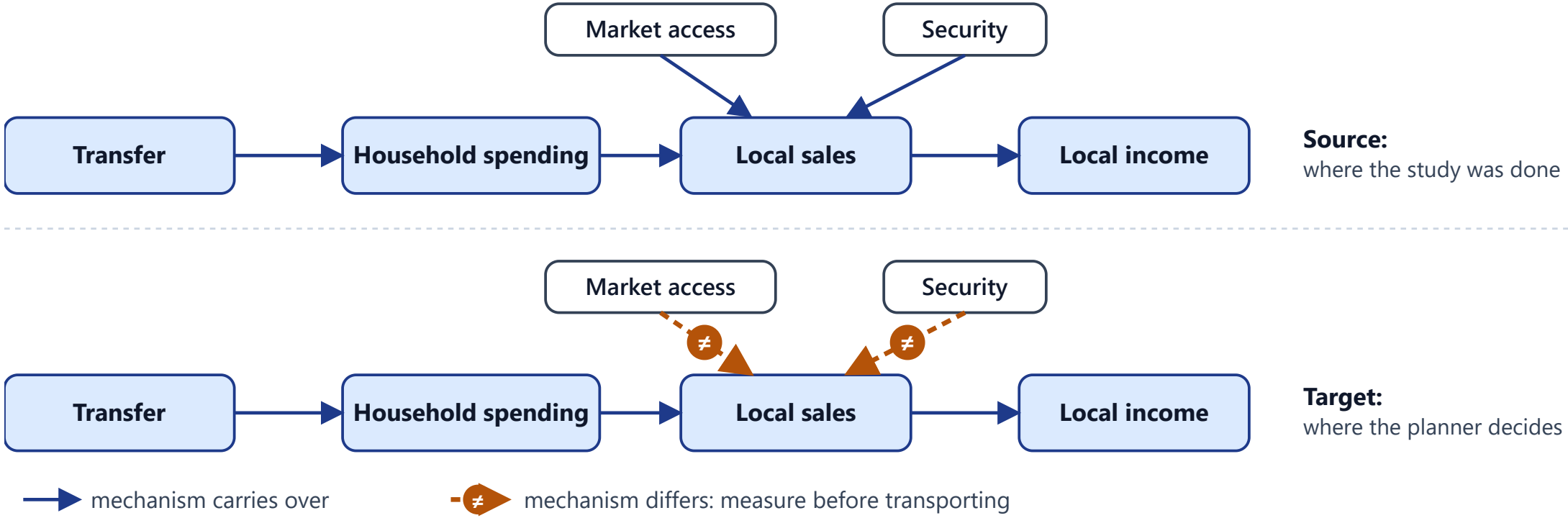


The planner and analyst work through six steps (left), repeating until the planner is satisfied. Step 5 returns one of four prediction types

Type	When
Point estimate with standard errors	Planner's and study's causal maps match, distance small
Bounds (Gechter)	Partial overlap
Ordinal category or sign	Only direction supported
"Insufficient evidence" or "distance too great"	No defensible transport: a result

Which differences matter: selection diagrams

A **selection diagram** is a causal graph, with boxes for variables and arrows for cause and effect, plus a mark on each arrow that works differently in the source and the target setting.



- Marked arrows say which differences **must be measured**, and whether a way to carry the result over exists at all
- Here the graph comes from the planner's decision map, matched to the source study's causal story

Approaches to explore, and the context data behind them

Group	Approach, in plain words	Where the expert enters
What can be carried over?	**Selection diagrams:** which causal links differ	The graph comes from the planner's map
How to combine evidence	**Hierarchical Bayesian models:** borrow strength across sites by estimating how much effects vary between them	Priors; which sites count as comparable
How to combine evidence	**Similarity-weighted borrowing:** weight studies by how close their setting is to the target	Which variables, and how close is close enough
What if it cannot?	**Partial-identification bounds:** report the range of effects consistent with what is known	The assumptions the bounds rest on

- **Context data:** remote sensing (rainfall, vegetation indices from Landsat, MODIS and Sentinel-2, night lights) plus conflict events, prices, displacement, boundaries
- **Limits:** satellites describe environment and activity, not treatment or welfare; resolution must suit the unit; site coordinates decide what can be linked
- **Exploratory test:** the Egger versus HSNP pair. The two report different quantities, so we first harmonize them, then ask whether the approaches agree on whether a result carried from one setting to the other

Timeline: three gates on the thesis schedule



What the gates are for: each is a go/no-go check before the next stage. Dates are the current plan. IRB is Columbia's ethics review for research with people.

- **Gate 1, answerable? (May 2027)** Faculty approve the audit and pre-registration: the question can be measured from documents
- **Gate 2, mappable? (Aug 2028)** Fieldwork shows practitioners' causal maps are usable
- **Gate 3, transportable? (Spring 2029)** The method is validated with a partner and written up

Appendix

A1. What a selection diagram can and cannot do

- It tells you which differences you **must measure** and whether a **formula to carry the result over exists at all**
- It **cannot** create data that do not exist, repair a missing outcome, or make an impossible delivery possible
- The formalism is Bareinboim and Pearl's; the graph here comes from the planner's decision map, matched to the source study's causal story

A2. The full decision table

Relevant evidence exists?	Discretion: yes	Discretion: no (fixed by rules, budget, mandate)
Yes: relevant and actionable	**Use it**	On the shelf
Yes, but too distant	**Gap: transport the evidence (evidence transport)**	Not a priority
Yes, but not actionable here	**Gap: redesign or adapt (decision mapping)**	Not a priority
None	**Gap: generate or substitute**	Ignore

- Scope: decisions where the planner already has discretion and lacks relevant evidence; evidence loosening constraints is plausible but out of scope
- Other states: little demand beyond logistics; abundant evidence not used; evidence used well; planners make do; insufficient evidence either way

A3. Comparability profile (illustration)

Dimension	Objective measure	Subjective rating (1 to 6, illustrative)
Environment	Overlap of the vegetation-index distributions	Planner: forest health 4
Market	Market-access or price-pass-through indicator	Planner: 3
Institution	Delivery channel (NGO, public safety net, payment rail)	Planner: 5
Conflict exposure	Event counts and trends	Planner: 2
Population	Displacement status, composition	Planner: 4

- Thresholds per dimension are set with the planner; a gap between the two columns on a row is flagged for discussion
- Values here are placeholders, not data

A4. Validation, gates and timeline

Research question	Validated when
Discretion and value of evidence	A working paper on the knowledge audit is approved by faculty, and at least one fit partner agrees with the funnel documented
Decision mapping	Desk phase: documents suffice to model constraints, scenarios and log-frames. Partner phase: the partner finds the process and products useful (needs ethics approval)
Evidence transport	Partial: a methodology paper with empirical validation. Full: a partner uses the results. Follow-on: predictions turn out reasonably correct

- **Dates (owner's plan, 2026-11-13):** faculty talk April 2027; faculty approval and pre-registration May 2027; IRB submission June 2027; recruitment fall 2027 to spring 2028; field work summer 2028; transport model validated with a partner fall 2028; thesis paper spring 2029
- **Case-level checks before investing in a case:** case and partner fit; is new evidence worth it; what can be inferred; can it be evaluated

A5. Bias controls for the feedback rounds

Someone will ask how planner feedback, with variables not fixed in advance, avoids p-hacking. The top ideas:

1. **A written, registered protocol:** the steps (data discovery, map matching, comparability profile, modelling, planner review, repeat), a stopping rule, fixed dimensions and approaches, and a log of every round and variable tried
2. **The planner rates similarity before seeing the predicted effect;** changes made after seeing the estimate are flagged
3. **Report sensitivity:** the prediction under every reasonable specification tried
4. **Lock the prediction before the outcome** (pre-registered, time-stamped)

Further controls (a planner-free baseline; more than one expert where available) are for the protocol to settle.

A6. Project maturity: built, planned, gated

Component	Status
Context panel (Somalia, Sudan)	Built and reproducing; separate config per country
`reprokit`	Released (v0.4.0, evidence kind added)
Literature discovery and extraction	Built and piloted
Evidence map (3ie-style gap map)	In design; study catalogue first
Bangladesh insurance replication	Nearly complete in full (owner, 2026-11-13); village coordinates are the gate for the satellite-linked step
Transferability	Planned; no repository yet
Data-fusion toolkit	Deferred

A7. Claims we keep, soften or drop

Claim	Treatment
"No one has audited evidence for programme design"	**Soften:** nearest prior art found
Egger versus HSNP as a transport failure	**Soften:** hypothesis-generating pair
A complete corpus of trials; universal impact ranges; a live simulator or dashboard	**Dropped**
Impact estimates in novel localities	**Soften** to a tiered claim
Participatory causal maps produce usable log-frames	**Proposal**
The compound-shocks panel supports transfer	**Narrowly:** context only
Evidence-use patterns across the field	**Qualitative only;** the scans were not systematic

A8. Positioning and dropped use cases

- **Prior art found:** Brown et al. 2025 and IDinsight (value of information for funding); ALNAP (immediate-response decisions); Elrha (barriers); Egger-style and Bedoya-style trials (transport and fragile-setting cases)
- **Dropped use cases:** anticipatory action and index insurance are examples of modality and timing questions only, not research strands of this agenda; operational logistics optimization is a separate adjacent agenda
- **Not a claim** about which intervention is best

A9. Open items to validate in the next few weeks

Item	Why	Next step
Scale figures (World Bank, UNHCR, OCHA)	Opening slide; found via a web-search summary	Gemini check against the primary reports; name the poverty line
Egger versus HSNP definitions	Egger's multiplier is GDP per unit transferred; HSNP's 0.38 to 0.93 may be *additional* income beyond the transfer (total about 1.38 to 1.93)	Read the HSNP local-economy methods; harmonize before any comparison
Replication feasibility	Egger's package is public but village locations are restricted; HSNP data held locally; Bedoya data not found	Confirm what each supports; ask authors only for what is needed
Bedoya et al.	Aspirational case; claims checked on the abstract and registration only	Full read; check what enabled success
Stage-2 extraction	The pipeline slide says it extracts causal claims; pilot figures not re-verified	Confirm against its status doc
Literature map (slide 4)	Row descriptions are my characterization	Read one paper per field
Evidence-use scans	The "mixed, sector-dependent" picture rests on about 31 sources I have not read	Read a sample before relying on it
Nexus definition	Mine, not sourced	Read the OECD DAC recommendation
Ethics and outreach	J-PAL contact could count as human-subjects research	Ask the IRB before anything beyond public-data requests

Design questions (decision taxonomy, headquarters/field relationship, the definition of "comparable") are tracked separately.

A10. Sources consulted (1 of 2): read in full or checked against the text

- **Programme documents:** FCDO, Kenya Hunger Safety Net Programme Phase 2 — Business Case and Completion Review
- **Egger, Haushofer, Miguel, Niehaus, Walker (2022).** General equilibrium effects of cash transfers: experimental evidence from Kenya. *Econometrica*
- **Bedoya, Coville, Haushofer, Isaqzadeh, Shapiro (2019).** No household left behind: Afghanistan Targeting the Ultra Poor impact evaluation. NBER WP 25981
- **Pritchett and Sandefur (2015).** Context matters for the size of impact: experiments in other settings (microcredit)
- **Hjort et al. (2021).** How research affects policy: Brazilian municipalities
- **Crost, Felter and Johnston (2014; 2016).** Aid, insurgent response and conflict, Philippines (community-driven development; conditional cash transfers)
- **Brown, Salisbury and Ball (2025).** Value of information applied to grant allocation
- **Groh et al. (2016); Tennant et al. (2021); Altmejd et al. (2019).** Checked against the transcripts for specific claims

Full-text transcripts are held locally (git-ignored); identifiers and DOIs are in the novelty-check notes.

A11. Sources consulted (2 of 2): identifier-matched, secondhand or not yet read

- **Matched to a registry, not read:** Vivalt (2020); Allcott (site selection bias); Meager (Bayesian hierarchical analysis of microcredit); Banerjee et al. (2016, proof of concept to scalable policies); Oliver et al. (2014, barriers to evidence use); Snilstveit et al. (2016, evidence gap maps); Kwakkel, Haasnoot and Walker (2016, robust decision-making and adaptive pathways); Bareinboim and Pearl (2012, transportability)
- **Read by Gemini only:** Gechter (bounds); Montiel Olea et al. (site selection); Bongers et al. (cyclic causal models); DellaVigna, Pope and Vivalt (2019); Kastner et al. (2009) and White et al. (2020) on capture-recapture; Bold et al. (2018); IDinsight (2022); ALNAP, Campbell and Knox Clarke (2019); Elrha, Smith, Swithern and Harrison (2022)
- **Two non-systematic scans of evidence use** (about 31 sources) by Gemini: not read by me; no figures used
- **Scale figures** (World Bank 2025; UNHCR Global Trends 2025; OCHA GHO 2026): web summaries, **being verified**
- **Not read:** OECD DAC nexus recommendation; Claxton (1999); Fenwick et al. (2008)

Reference hygiene: identifiers come from registries (Crossref), not model recall.

A12. Why it matters beyond this case

- **Development economics:** context-dependent external validity of causal treatment effects for interventions
- **Decision science:** value of information applied to programme-design decisions, with measured discretion
- **Causal inference:** transport methods tested prospectively, with pre-registered predictions, in a hard setting
- **Climate and environmental decision-making:** a decision-relevance test for adaptation methods under uncertainty
- **Research funding and synthesis:** separating real research gaps from gaps existing evidence can fill
- **Research infrastructure:** an open-source, reproducible pipeline others can re-run on a different decision
- **Humanitarian practice:** a map of what informs plans, and where central rules block use of evidence

Intended contributions, not results. Each depends on the corresponding question working.